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Substations and Lines (MPA-ESV 25/26L)

MPA - MESV Project 1

As an engineer in an engineering company you have to design a new earthing system for defined site (Secondary substation, HV tower etc., see Table with assignment variants). The customer (the distribution company) have provided you the necessary information (listed in the table). A 4-point Wenner measurement of soil resistivity was conducted by your company at the earthing system location site with electrode spacing 1, 3 and 5 meters. The designed earthing system needs to be compliant to permissible touch voltages, i.e. $U_{vT} < U_{vTp}$. The earthing system consists of two peripheral earthing electrode (i.e. two rectangular electrodes). The dimensions of the first rectangular electrode are included in the table. The mutual coefficient of the two rectangular electrodes is 0.69. The public is not expected to wear shoes in the earthing system vicinity. Earthing system is made of a zinc alloy steel FeZn strip of cross section 30x4 mm.

- 1) Determine permissible touch voltage U_{vTp} .
- 2) Determine soil resistivity for earthing system calculation ρ_E .
- 3) Determine earth fault current value I_E .
- 4) Determine earth resistance of the first earthing electrode (i.e. first rectangular electrode) R_1 .
- 5) Determine earth resistance of the second earthing electrode R_2 .
- 6) Determine earth resistance of combination of both electrodes R_{12} .
- 7) Evaluate compliance with permissible touch voltage condition, $U_{vT} < U_{vTp}$.
- 8) Include simplified drawing of your design including its dimensions.
- 9) In the project include conclusion.

Name Surname	ID	$I_f(A)$	$t(s)$	Star point connection	Voltage level	Line type	Eart wire/ Cable stealth	Reduction factor (r)	ES Type	Shoes	R 1m	R 3m	R 5m	A m	B m	Depth m
Mirjabbar Badalov	285269	250	0.8	Resistor Earthed	22kV	Cable	Yes	0.2	Secondary Sub	No	57	78	47	2	2	0.6

1) Determine permissible touch voltage U_{vTp} :

The permissible touch voltage (U_{vTp}) is dependent on the fault duration ($t_f = 0.8$ s) and body impedance characteristics. Based on the standard EN 50522 Bieglmeyer curves for a fault clearance time of 0.8 seconds, the prospective permissible touch voltage for a human body is approximately 105 V.

Because the public is not expected to wear shoes in the vicinity, we do not add any additional series shoe resistance

$$(RF1 = 0). U_{vTp} \approx 105 \text{ V}$$

2) Determine soil resistivity for earthing calculation ρ_E :

$$\rho_E = (57 + 78 + 47) / 3 = 60.67 \text{ } \Omega \cdot \text{m}$$

3) Determine earth fault current value I_E :

- $I_E = r * I_F$
- $I_E = 0.2 * 250 \text{ A} = 50 \text{ A}$

4) Determine earth resistance of the first earthing electrode R_1 :

- Area (S_1): $2 \text{ m} * 2 \text{ m} = 4 \text{ m}^2$
- Equivalent diameter of grid (D_{eq1}): $D_{eq1} = \sqrt{(4 * S_1 / \pi)} = \sqrt{(16 / 3.1415)} = 2.257 \text{ m}$
- Equivalent diameter of 30x4mm strip (d): For a 30x4 mm strip, the standard half-width approximation gives $d = 0.015 \text{ m}$.

Using the ring resistance formula:

- $R_1 = \{ \rho_E / [2 * \pi^2 * D_{eq1}] \} * [\ln(8 * D_{eq1} / d) + \ln(\pi * D_{eq1} / (2 * z))]$
- $R_1 = \{ 60.67 / [2 * \pi^2 * 2.257] \} * [\ln(8 * 2.257 / 0.015) + \ln(\pi * 2.257 / (2 * 0.6))]$
- $R_1 = 1.362 * [\ln(1203.7) + \ln(5.909)]$
- $R_1 = 1.362 * [7.093 + 1.776] = 12.08 \text{ } \Omega$

5) Determine earth resistance of the second earthing electrode R_2 :

- **Area (S_2):** $4 \text{ m} * 4 \text{ m} = 16 \text{ m}^2$
- **Equivalent diameter of outer grid (D_{eq2}):** $D_{eq2} = \sqrt{(4 * 16 / \pi)} = 4.513 \text{ m}$
- $R_2 = \{ 60.67 / [2 * \pi^2 * 4.513] \} * [\ln(8 * 4.513 / 0.015) + \ln(\pi * 4.513 / 1.2)]$
- $R_2 = 0.681 * [\ln(2406.9) + \ln(11.815)]$
- $R_2 = 0.681 * [7.786 + 2.469] = 6.98 \Omega$

6) Determine earth resistance of combination of both electrodes R_{12} :

The two electrodes are connected in parallel, with the mutual interference coefficient given as $\eta_{12} = 0.69$.

- $R_{12} = [(R_1 * R_2) / (R_1 + R_2)] * (1 / \eta_{12})$
- $R_{12} = [(12.08 * 6.98) / (12.08 + 6.98)] * (1 / 0.69)$
- $R_{12} = [84.32 / 19.06] * 1.449$
- $R_{12} = 4.424 * 1.449 = 6.41 \Omega$

7) Evaluate compliance with permissible touch voltage condition ($U_{vT} < U_{vTp}$)

First, calculate the total Earth Potential Rise (EPR), denoted as U_{epR} :

- $U_{epR} = I_E * R_{12} = 50 \text{ A} * 6.41 \Omega = 320.5 \text{ V}$

Using the quick safety design check approximation for the prospective touch voltage (U_{vT}):

- $U_{vT} \approx U_{epR} / 2$
- $U_{vT} \approx 320.5 \text{ V} / 2 = 160.25 \text{ V}$

Evaluation: U_{vT} (160.25 V) is **greater than** U_{vTp} (105 V). **Condition Failed:** 160.25 V is not less than 105 V. The design is NOT compliant with safety thresholds.

The earthing system for the secondary substation was systematically designed and calculated

1. Applying the cable sheath reduction factor (0.2), the effective fault current (I_E)

- Final Recommendation:** The earthing design does **not comply** with the safety target (U_{VT} must be less than U_{VTp}). The engineering design must be further expanded. Adding a third concentric outer ring or incorporating deep earthing rods are required steps to bring the touch voltage down to the acceptable 105 V threshold.